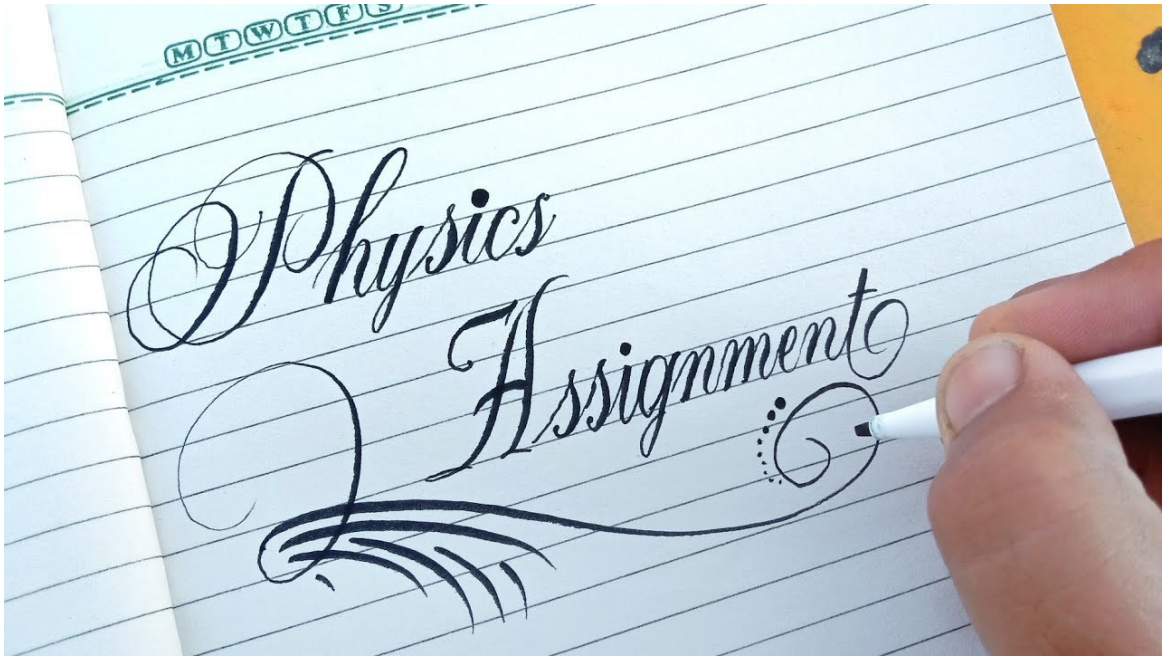
XIAMEN UNIVERSITY MALAYSIA

ASSIGNMENT 1



Lecture1-5

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Try to solve the questions **Completely with all steps**! (Effective for grading)

Only **handwriting** will be accepted (no typing please!).

The answer can be in **ONE PDF** format file only (individual activity)

Cheating in any condition will **consider a zero Mark** in any exam and assignment!

only **1 Attempt** is allowed.

The deadline is the **1st of November 2022 at 11:00 AM (Morning Time)**

Late submission will impose **a penalty (communicate with me if a delay happened)**!

Cheating and writing from **Internet resources** will consider cheating, please avoid any kind of cheating! (**Do not share answers** at any cost!)

Please use the **Assignment template**, otherwise, your script will not be graded.

1. The siren of a fire engine that is driving northward at 30.0 m/s emits a sound of frequency 2000 Hz. A truck in front of this fire engine is moving northward at 20.0 m/s. (a) What is the frequency of the siren's sound that the fire engine's driver hears reflected from the back of the truck? (b) What wavelength would this driver measure for these reflected sound waves?

2. A point charge is placed at each corner of a square with side length a. All charges have magnitude q. Two of the charges are positive and two are negative (Fig. A1.1). What is the direction of the net electric field at the center of the square due to the four charges, and what is its magnitude in terms of q and a?

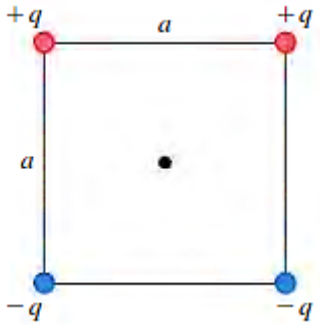


Figure A1.1

3. Adjacent antinodes of a standing wave on a string are 15.0 cm apart. A particle at an antinode oscillates in simple harmonic motion with an amplitude of 0.850 cm and a period of 0.0750 s. The string lies along the + x-axis and is fixed at $x = 0$. (a) How far apart are the adjacent nodes? (b) What are the wavelength, amplitude, and speed of the

two traveling waves that form this pattern? (c) Find the maximum and minimum transverse speeds of a point at an antinode. (d) What is the shortest distance along the string between a node and an antinode?

4. If two electrons are each 1.50×10^{-10} m from a proton (Fig. A1.2), find the magnitude and direction of the net electric force they will exert on the proton.

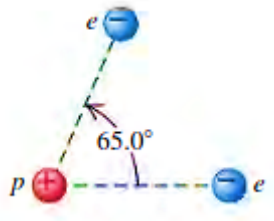


Figure A1.2

5. A firework charge is detonated many meters above the ground. At a distance of d_1= 500 m from the explosion, the acoustic pressure reaches a maximum of DP_{max} = 10.0 Pa (Figure A1.3). Assume the speed of sound is constant at 343 m/s throughout the atmosphere over the region considered, the ground absorbs all the sound falling on it, and the air absorbs sound energy as described by the rate of 7.00 dB/km. What is the sound level (in decibels) at a distance of $d_2 = 4.00 \times 10^3$ m from the explosion? (ρ=1.2)

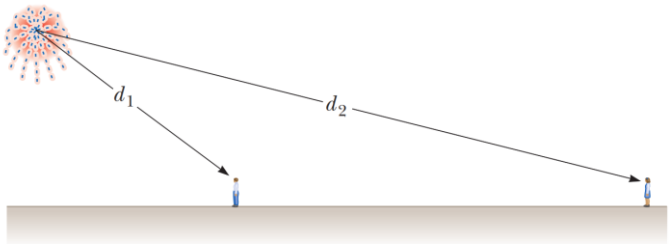


Figure A1.3

6. A very light rigid rod of length 0.500 m extends straight out from one end of a meterstick. The combination is suspended from a pivot at the upper end of the rod as shown in Figure A1.4. The combination is then pulled out by a small angle and released.

(a) Determine the period of oscillation of the system.

(b) By what percentage does the period differ from the period of a simple pendulum 1.00 m long?

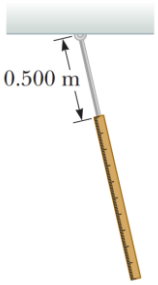


Figure A1.4

7. Liu is studying insulator materials, now he is considering having a thin insulating three-quarter circle of radius, R, y is held fixed in the XY-plane with its center at the origin of the coordinate system, as he draws in Figure A1.5. It carries a negative charge of -3Q uniformly distributed over its length. An electron is released from rest at the origin.

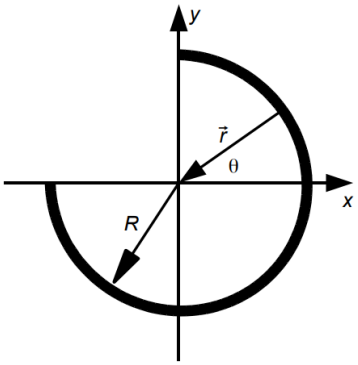


Figure A1.5

(a) Find the initial acceleration of the electron in terms of the system parameters. Express your answer in unit vector notation (Gravity may be neglected).

(b) Determine the maximum speed the electron reaches in terms of the system parameters.

8. Cheng is interested to study electric charges with different setups or configurations. According to his findings in 1 dimension, the calculation shall be simple however he wants to design the 3 dimensions structure to calculate the charge accordingly so that he put eight charged particles, each of magnitude q, located on the corners of a cube of edge **s** as shown in Figure A1.6. (a) Determine the x, y, and z components of the total force exerted by the other charges on the charge located at point A. What are (b) the magnitude and (c) the direction of this total force?

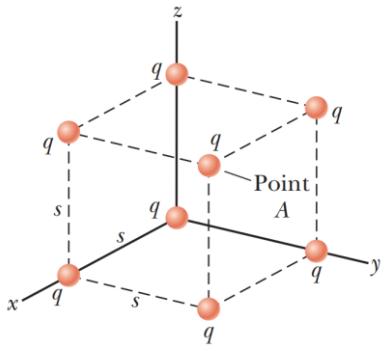


Figure A1.6

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